DP Biology – A2.3 (HL only)

Viruses

- How do viruses manage with so few genes?
- How do viruses differ from one another?

HL statements A2.3.1–6. Textbook pp. 88–99.

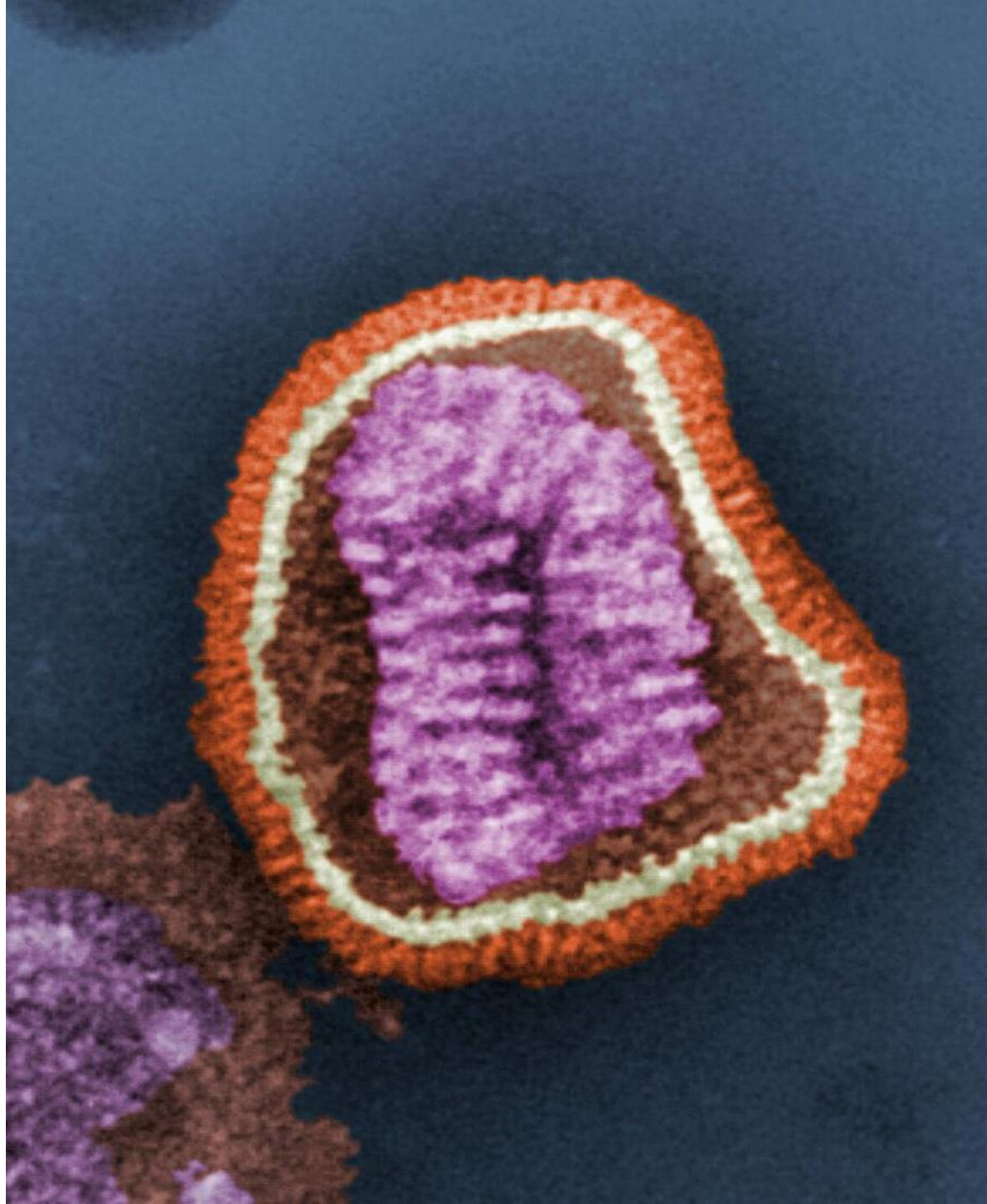


Image: an influenza virus particle (TEM, coloured). C. Goldsmith, CDC, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	A2.3.1–2	Describe the features all viruses share, and the diversity of their genomes, capsids and envelopes
2	A2.3.3–4	Compare the lytic and lysogenic cycles of bacteriophage lambda
3	A2.3.5–6	Evaluate hypotheses for the origins of viruses, and explain rapid evolution in influenza and HIV

The big picture: borrowed machinery



A virus is a **genome in a protein coat**. It has no metabolism, so it uses a host cell's ribosomes, enzymes and ATP.

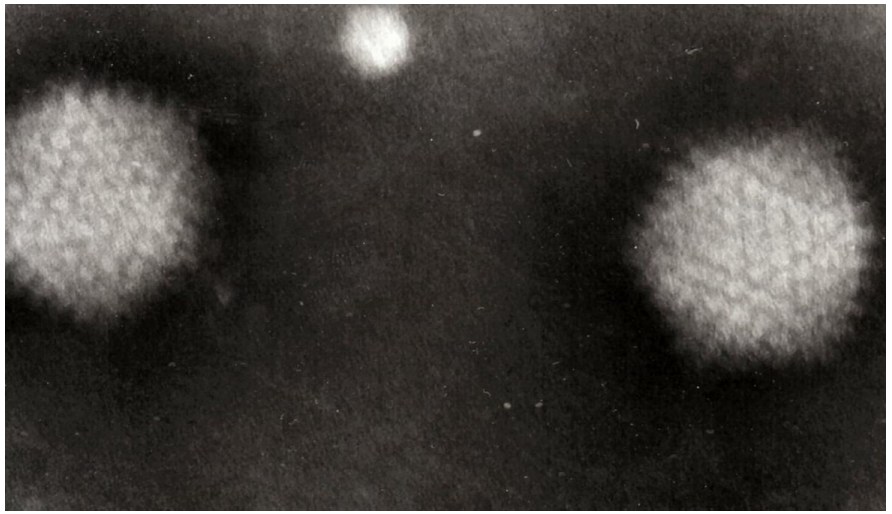
Big question

Why do you need a new flu vaccine every year, but only one measles vaccine for life?

Features shared by all viruses

A2.3.1

Feature	Detail
Small, fixed size	About 20–400 nm; assembled, not grown
Nucleic acid	DNA or RNA, never both
Protein capsid	Made of capsomeres
No cytoplasm	No ribosomes or metabolism
Few enzymes	Use the host's



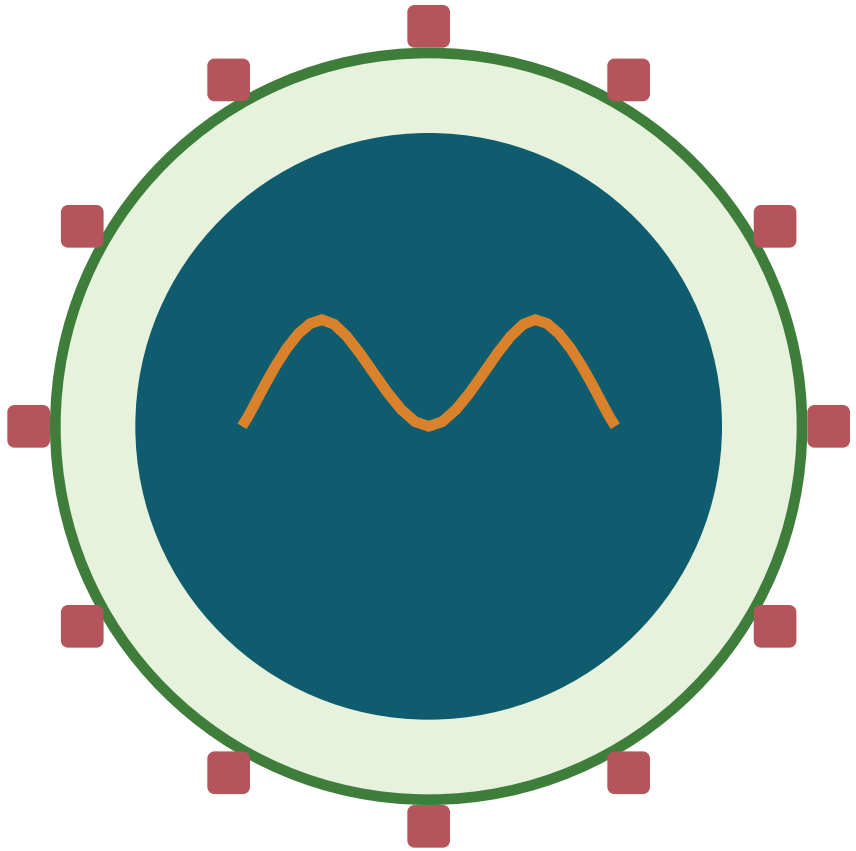
Adenovirus particles (TEM). GrahamColm, CC BY 3.0.

Common mistake

A virus is not a cell. It is an obligate parasite: it reproduces only inside a host.

Anatomy of a virus

A2.3.1–2



glycoproteins (spikes)

envelope (host membrane)

capsid (protein)

genome: DNA or RNA

Not all viruses

Only enveloped viruses (HIV, flu) have the outer membrane and spikes.

Viruses compared with bacteria

A2.3.1

	Virus	Bacterium
Cellular?	No	Yes
Diameter	0.02–0.4 μm	1–5 μm
Nucleic acid	DNA or RNA	Both
Ribosomes	None	70S
Reproduction	Assembled in a host	Binary fission

Viruses were predicted before being seen: filtered sap still caused disease.

Diversity of viral genomes

A2.3.2

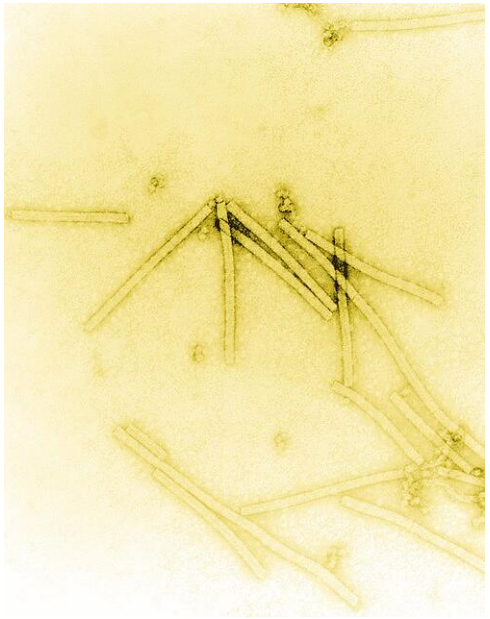
Genome	Example	Note
Double-stranded DNA	Phage lambda	Circularizes in the host
Single-stranded RNA (+)	SARS-CoV-2	Acts directly as mRNA
Single-stranded RNA (-)	Influenza	Copied to mRNA first; 8 segments
RNA, retrovirus	HIV	Reverse transcriptase makes DNA

Key idea

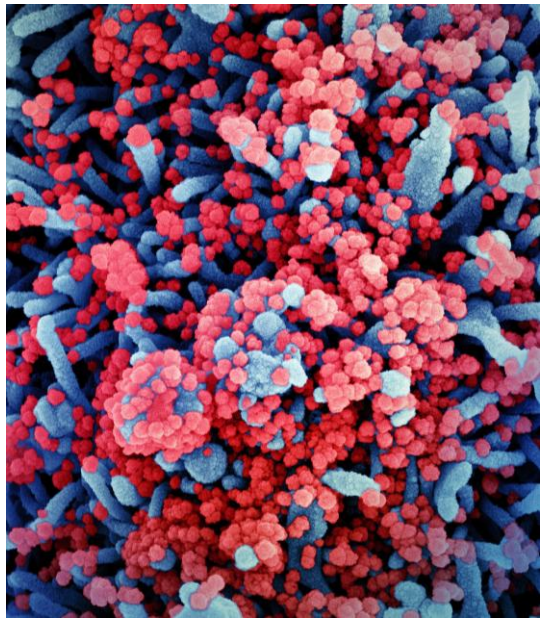
DNA or RNA; single or double; linear, circular or segmented.

Capsids and envelopes

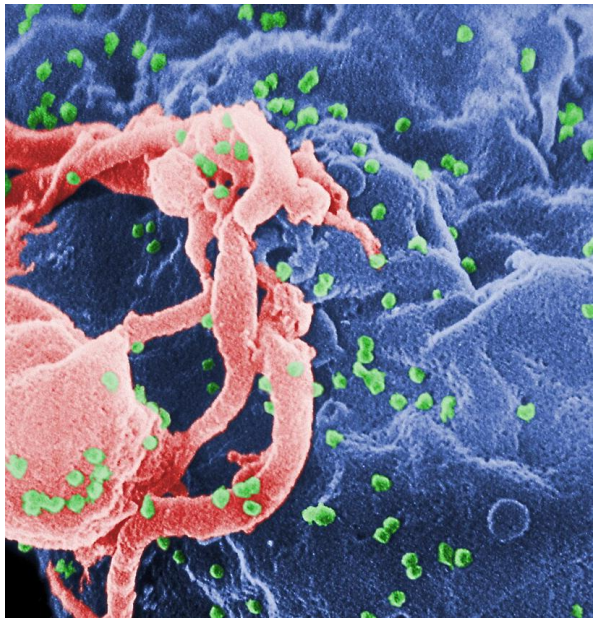
A2.3.2



TMV rods (TEM). Public domain.



SARS-CoV-2 on a cell (SEM). NIH, public domain.



HIV budding from a cell. CDC, public domain.

Capsids: **helical**, **icosahedral** or **complex**.

An **envelope** is host membrane with viral proteins.

	Enveloped	Not enveloped
Examples	HIV, flu, coronaviruses	Lambda, TMV, adenovirus
Leaves by	Budding	Cell bursting

Comparing three viruses

A2.3.2

	Phage lambda	SARS-CoV-2	HIV
Nucleic acid			
Strands			
Envelope?			
Host cell			

Activity

Complete in pairs. Which feature most limits each virus's host range?

Check-in: class 1

A2.3.1–2

1. List three features shared by all viruses.
2. Give three differences between a virus and a bacterium.
3. Give two ways viral genomes vary.
4. Distinguish enveloped and non-enveloped viruses.
5. Name three capsid shapes.

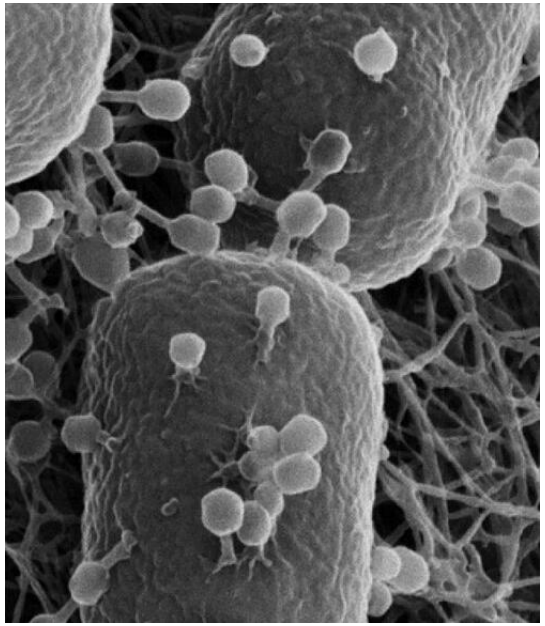
How to use this

Answer without your notes, then check against slides 3–9.

The lytic cycle of phage lambda

A2.3.3

Attach: tail binds E. coli → Inject DNA; capsid stays out → Replicate: host enzymes copy → Assemble: host ribosomes make coats → Lysis: about 100 phages released

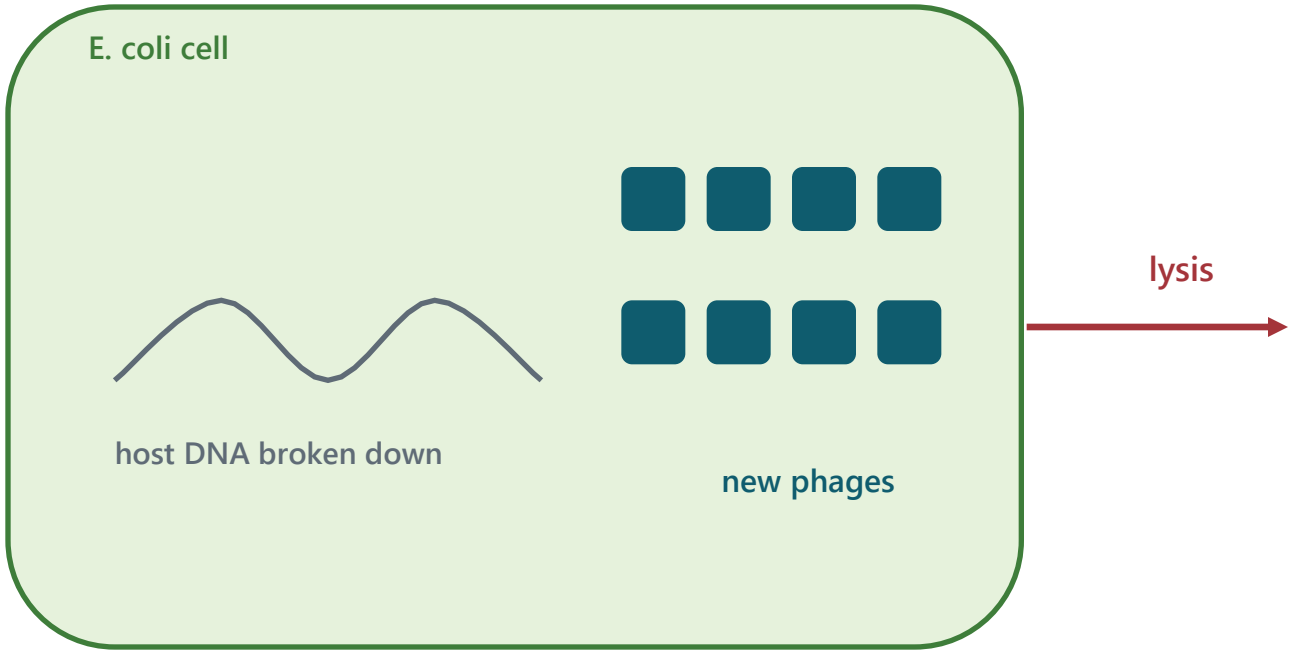


E. coli with phages attached. McGenity et al., CC BY-SA 4.0.

Using the host

The phage uses the host's ATP, nucleotides, amino acids, enzymes and ribosomes. The host cell dies.

Inside the host



Phage genes take over: the host builds **phage DNA and proteins**, then **lysozyme-like enzymes** burst the wall.

Virulent

Phages that use only this cycle are called virulent.

The lysogenic cycle

A2.3.4



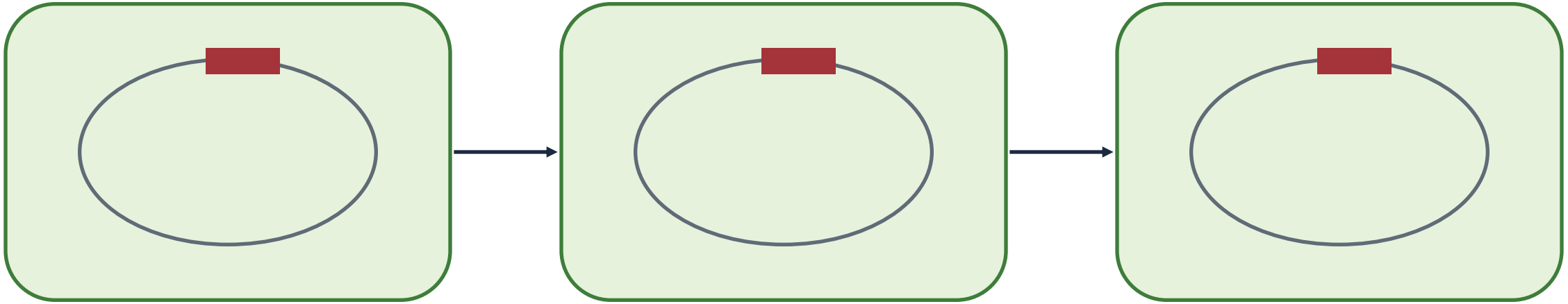
The phage **hides** in the host genome. The host survives and passes the prophage to every daughter cell.

Why switch?

When the host is stressed or dying, leaving by the lytic cycle saves the virus.

A hidden passenger

A2.3.4



red: prophage in the circular chromosome, copied into every daughter cell

Think

Why might a lysogenic phage spread further in a population of slowly growing bacteria?

Lytic and lysogenic compared

A2.3.3–4

	Lytic	Lysogenic
Phage DNA	Separate	Integrated (prophage)
Host cell	Killed quickly	Survives and divides
New viruses	Many at once	None until induced
Favoured when	Bacteria grow well	Bacteria grow poorly

Exam tip

Name the stages in order and say which host machinery each uses.

Which cycle is better for the virus?

A2.3.4

	Advantages	Disadvantages
Lytic		
Lysogenic		

Hint

Think about speed, numbers, surviving when hosts are scarce, and spreading by host division.

Check-in: class 2

A2.3.3–4

1. Outline the five stages of the lytic cycle.
2. State what a prophage is.
3. Explain how a prophage spreads.
4. Give a trigger for leaving the lysogenic cycle.
5. Compare the cycles in two ways.

How to use this

Answer without your notes, then check against slides 11–16.

Where did viruses come from?

A2.3.5

Progressive

Escaped mobile genes gained a coat

Regressive

Parasitic cells lost genes

Virus first

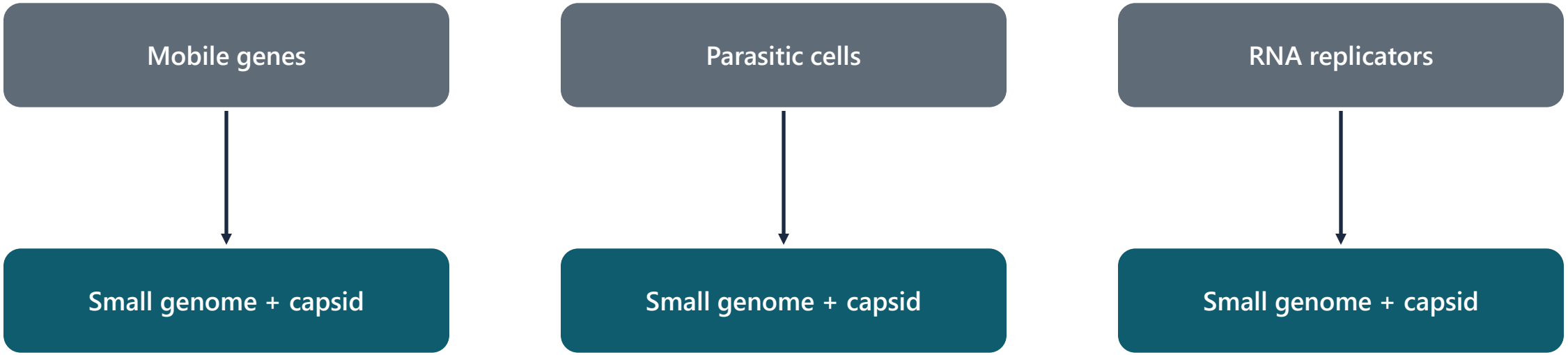
From the RNA world, before cells

Evidence for several origins

No gene is shared by all viruses, so probably no single ancestor. Shared features are likely convergent: similar adaptations to parasitism. Viruses use the same genetic code as their hosts.

Convergent evolution of viruses

A2.3.5



Key idea

Different starting points, same way of life, similar features: like wings in birds and bats.

Why some viruses evolve so fast

A2.3.6

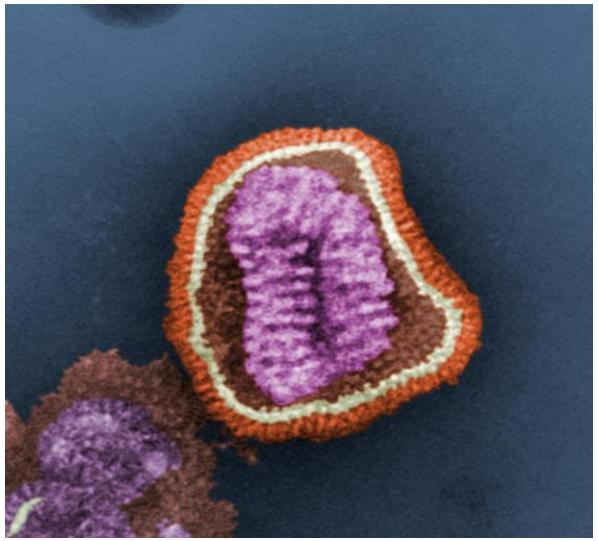
Factor	Effect
Short generation time	Many generations a year
Huge numbers	Rare mutations occur often
No proofreading	RNA replicase and reverse transcriptase make errors
Strong selection	Immunity and drugs select escape variants

Consequence

Vaccines and drugs lose their effect; flu vaccines change yearly.

Influenza: drift and shift

A2.3.6



Influenza virus (TEM). CDC, public domain.

Drift: small mutations in H and N

Immunity slowly fades: yearly vaccines

Shift: strains swap whole segments

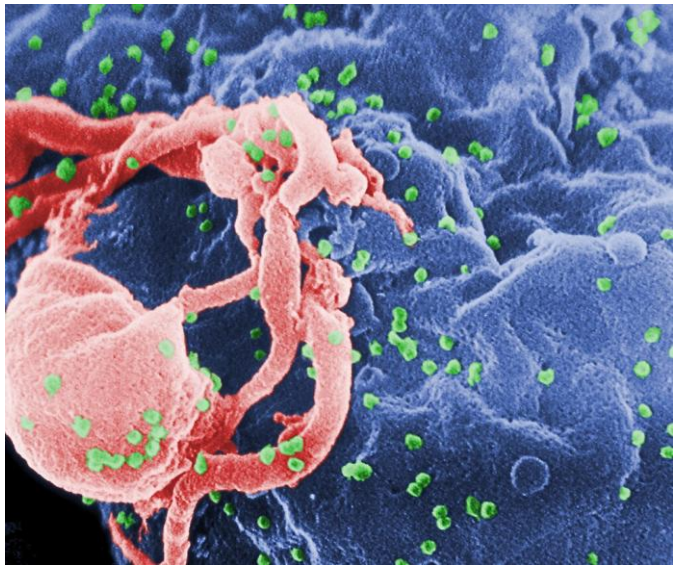
A new subtype: pandemic risk

Pandemics

Pigs and birds can carry several strains, so new combinations arise there. 1918 and 2009 pandemics were new influenza A subtypes.

Rapid evolution: HIV

A2.3.6



HIV budding (SEM). CDC, public domain.

Reverse transcriptase has no proofreading, so most new viruses carry a mutation.

One person can carry many variants within years.

Treatment

Drug-resistant variants are selected quickly, so therapy combines three or more drugs. Changing envelope proteins make a vaccine very hard.

Working with data: flu vaccine effectiveness

A2.3.6

Seasonal flu vaccine effectiveness, five seasons (rounded values in the range reported by health agencies).

Season	Effectiveness / %
A	60
B	19
C	48
D	39
E	54

1. Calculate the mean.
2. Suggest why season B was low.
3. Explain why a new vaccine is needed yearly.

Check-in: class 3

A2.3.5–6

1. Outline three hypotheses for the origin of viruses.
2. Explain why viruses probably have several origins.
3. Give three reasons some viruses evolve fast.
4. Distinguish antigenic drift and shift.
5. Explain why HIV is treated with drug combinations.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
Viruses are tiny bacteria.	They are not cells.
All viruses have DNA.	Some have RNA.
Antibiotics kill viruses.	They act on bacteria only.
Lysogenic hosts die at once.	The prophage is copied quietly.

Summary: viruses

A2.3.1–6

	Key idea
A2.3.1	Shared: small, genome, capsid, no metabolism
A2.3.2	Diverse genomes, capsids, envelopes
A2.3.3	Lytic: attach, inject, replicate, assemble, lyse
A2.3.4	Lysogenic: prophage copied with host
A2.3.5	Several origins; convergent features
A2.3.6	Fast evolution: flu drift/shift, HIV

Check that you can...

1. **List** features shared by all viruses
2. **Outline** diversity of viral structure
3. **Compare** lytic and lysogenic cycles
4. **Evaluate** hypotheses for viral origins
5. **Explain** rapid evolution in influenza and HIV

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State three features that all viruses share.

2. Distinguish between the lytic and lysogenic cycles.

3. Explain why influenza vaccines are changed every year.

Image credits

Influenza virus particle color.jpg – Photo Credit: Cynthia Goldsmith Content Providers(s): CDC/ D – Public domain

Adenovirus 4.jpg – GrahamColm – CC BY 3.0

TobaccoMosaicVirus.jpg – Unknown author – Public domain

Novel Coronavirus SARS-CoV-2 (50098378372).jpg – NIH Image Gallery from Bethesda, Maryland, USA – Public domain

HIV-budding-Color.jpg – Photo Credit: C. Goldsmith Content Providers: CDC/ C. Goldsm – Public domain

Escherichia coli with phages.jpg – Terry J. McGenity, Amare Gessesse, John E. Hallsworth, Esthe – CC BY-SA 4.0

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